

Predicting local lithium-plating onset with upscaled models: supplementary material

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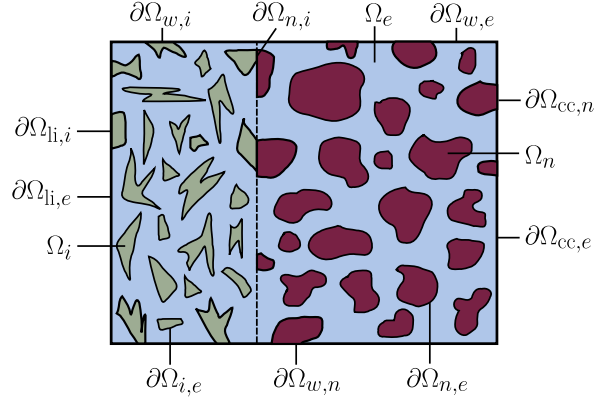


Figure 12: A schematic of the half cell modelled.

1 The Microscale Model

A schematic of the half-cell modelled is shown in Figure 12. As in the main text, we refer to the working electrode as ‘negative.’ This is not technically accurate, since in a half cell the electrode would in fact be positive relative to a lithium foil. Nonetheless we maintain this convention to avoid confusion, as we are interested in the negative electrode of a full cell.

1.1 Mass and Charge Conservation

Overall, two balance equations are required in each phase – a mass and charge balance, each accompanied by a boundary condition (BC) at the internal interface to describe the electrochemical reaction. The system of equations for mass conservation in the negative electrode active material (AM) and electrolyte is summarised below,

$$\partial_t c_e + \nabla \cdot \mathbf{N}_e = 0 \quad \text{in } \Omega_e, \quad (49a)$$

$$\partial_t c_n + \nabla \cdot \mathbf{N}_n = 0 \quad \text{in } \Omega_n, \quad (49b)$$

$$-\mathbf{n} \cdot \mathbf{N}_e = F^{-1} j_n \quad \text{at } \partial\Omega_{n,e}, \quad (49c)$$

$$-\mathbf{n} \cdot \mathbf{N}_n = F^{-1} j_n \quad \text{at } \partial\Omega_{n,e}, \quad (49d)$$

where c_n and c_e denote lithium/lithium-ion concentration in the negative electrode AM and electrolyte respectively. On the other hand, conservation of charge gives,

$$\nabla \cdot \mathbf{i}_e = 0 \quad \text{in } \Omega_e, \quad (50a)$$

$$\nabla \cdot \mathbf{i}_n = 0 \quad \text{in } \Omega_n, \quad (50b)$$

$$-\mathbf{n} \cdot \mathbf{i}_e = j_n \quad \text{at } \partial\Omega_{n,e}, \quad (50c)$$

$$-\mathbf{n} \cdot \mathbf{i}_n = j_n \quad \text{at } \partial\Omega_{n,e}, \quad (50d)$$

where F is Faraday’s constant. In Figure 12, the solid inactive material domain, Ω_i , is shown. No mass or charge transport exists in Ω_i , thus no conservation equations need be solved. The BCs at the AM-inactive ($\partial\Omega_{n,i}$) and electrolyte-inactive ($\partial\Omega_{i,e}$) interfaces are,

$$\mathbf{n} \cdot \mathbf{N}_n = 0 \quad \text{at } \partial\Omega_{n,i},$$

$$\mathbf{n} \cdot \mathbf{i}_n = 0 \quad \text{at } \partial\Omega_{n,i},$$

$$\mathbf{n} \cdot \mathbf{N}_e = 0 \quad \text{at } \partial\Omega_{i,e},$$

$$\mathbf{n} \cdot \mathbf{i}_e = 0 \quad \text{at } \partial\Omega_{i,e}.$$

Note that this not relevant for the Results I Section, where no inactive material is present. To complete the model above, expressions for $\mathbf{N}_e$, $\mathbf{N}_n$, $\mathbf{i}_e$, $\mathbf{i}_n$, and j_n are needed. The electrolyte flux and current density equations are provided by concentrated-solution theory. Mass transport within the AM phase is modelled by Fickian diffusion, and charge transport by Ohm's law,

$$\mathbf{N}_e = -D_e \nabla c_e + \frac{t_+^0}{F} \mathbf{i}_e, \quad (52a)$$

$$\mathbf{i}_e = -\sigma_e \nabla \phi_e - \sigma_e \theta \nabla \ln c_e, \quad (52b)$$

$$\mathbf{N}_n = -D_n \nabla c_n, \quad (52c)$$

$$\mathbf{i}_n = -\sigma_n \nabla \phi_n, \quad (52d)$$

where c_e , ϕ_e , D_e , σ_e , and t_+^0 are the electrolyte concentration, potential, diffusivity, conductivity, and transference number (respectively). The AM potential and conductivity are ϕ_n and σ_n respectively, whilst

$$\theta = -\frac{2RT}{F} \frac{(1 - t_+^0)}{(1 + \delta_e)}, \quad (53)$$

where R is the gas constant, T is temperature, and $(1 + \delta_e)$ is the thermodynamic factor. The reaction rate, j_n , is assumed to follow a Butler-Volmer expression,

$$j_n(\phi_e, \phi_n, c_e, c_n) = 2j_{n0} \sinh\left(\frac{F}{2RT} \eta_n\right), \quad (54)$$

where

$$\eta_n = \phi_n - \phi_e - U_n(c_n), \quad (55)$$

$$j_{n0} = Fk_n \sqrt{c_n c_e (c_{n,\max} - c_n)}, \quad (56)$$

and j_{n0} is the exchange current density, η_n is the reaction overpotential, U_n is the open-circuit potential (OCP), k_n is the rate constant, and $c_{n,\max}$ is the maximum concentration of lithium in the AM. The above choices for describing transport and kinetics have become a standard approach to microscale modelling in the literature [14, 15, 17, 30, 37, 53].

1.2 External BCs

Finally, external BCs are required to complete the model. Besides the interface between AM and electrolyte ($\partial\Omega_{n,e}$), the two phases meet the current collector ($\partial\Omega_{cc,e}$ and $\partial\Omega_{cc,n}$) and outer walls ($\partial\Omega_{w,e}$ and $\partial\Omega_{w,n}$). Additionally, the half cell contains a separator region, and an interface exists between the lithium foil and electrolyte in the separator ($\partial\Omega_{li,e}$). For mass conservation,

$$\mathbf{n}_{e,w} \cdot \mathbf{N}_e = 0 \quad \text{at } \partial\Omega_{w,e}, \quad (57a)$$

$$\mathbf{n}_{n,w} \cdot \mathbf{N}_n = 0 \quad \text{at } \partial\Omega_{w,n}, \quad (57b)$$

$$\mathbf{n}_{e,cc} \cdot \mathbf{N}_e = 0 \quad \text{at } \partial\Omega_{cc,e}, \quad (57c)$$

$$\mathbf{n}_{n,cc} \cdot \mathbf{N}_n = 0 \quad \text{at } \partial\Omega_{cc,n}, \quad (57d)$$

$$\mathbf{n}_{e,li} \cdot \mathbf{N}_e = -\frac{j_{li}}{F} \quad \text{at } \partial\Omega_{li,e}, \quad (57e)$$

where $\mathbf{n}_{\alpha,\beta}$ represents the normal vector pointing outward from phase α to phase β , i_{app} is the current density applied to the current collector, j_{li} is the reaction rate at the lithium foil,

$$j_{li} = 2j_{li,0} \sinh\left(-\frac{F}{2RT} \phi_e\right), \quad (58)$$

and $j_{\text{li},0}$ is the exchange current density for the lithium foil. For charge conservation,

$$\mathbf{n}_{e,w} \cdot \mathbf{i}_e = 0 \quad \text{at } \partial\Omega_{w,e}, \quad (59a)$$

$$\mathbf{n}_{n,w} \cdot \mathbf{i}_n = 0 \quad \text{at } \partial\Omega_{w,n}, \quad (59b)$$

$$\mathbf{n}_{e,cc} \cdot \mathbf{i}_e = 0 \quad \text{at } \partial\Omega_{cc,e}, \quad (59c)$$

$$\mathbf{n}_{n,cc} \cdot \mathbf{i}_n = i_{\text{app}} \quad \text{at } \partial\Omega_{cc,n}, \quad (59d)$$

$$\mathbf{n}_{e,\text{li}} \cdot \mathbf{i}_e = -j_{\text{li}} \quad \text{at } \partial\Omega_{\text{li},e}. \quad (59e)$$

2 The Upscaled Models

In this section we give the model equations for both upscaled formulations: the DFN and the DC model. The principal difference between the two models is the treatment of the AM phase. In the DFN, a microscale description of the AM is maintained, albeit simplified by assuming that AM particles are isolated spheres. In the DC model, the AM phase is upscaled, using a closure problem to map between volume-averaged concentration and surface concentration. For a deeper discussion on the model similarities and differences, see our previous work [18, 19]. The formulation described below is for a half cell.

2.1 Reaction Rate and Surface Concentration

The way in which j_n is calculated gives rise to a key difference between the DFN and DC model. They both use the Butler-Volmer equation,

$$j_n^m = 2j_{n0}^m \sinh\left(\frac{F}{2RT}\eta_n^m\right), \quad (60)$$

where $m \in \{\text{dfn}, \text{dc}\}$ and

$$\eta_n^m = \langle \phi_n \rangle^n - \langle \phi_e \rangle^e - U_n(\langle c_n \rangle_A^m), \quad (61a)$$

$$j_{n0}^m = Fk_n \sqrt{\langle c_n \rangle_A^m \langle c_e \rangle^e (c_{n,\text{max}} - \langle c_n \rangle_A^m)}. \quad (61b)$$

They differ only in their estimate of surface concentration, $\langle c_n \rangle_A^m$. In the DFN, the estimate of surface concentration comes from the solution of the material balance in the representative sphere, $\langle c_n \rangle_A^{\text{dfn}} = c_n|_{R_n}$,

$$\partial_t c_n = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 D_n \frac{\partial c_n}{\partial r} \right) \quad \text{for } 0 \leq r \leq R_n, \quad (62a)$$

$$-D_n \frac{\partial c_n}{\partial r} = \frac{j_n^{\text{dfn}}}{F} \quad \text{at } r = R_n, \quad (62b)$$

$$-D_n \frac{\partial c_n}{\partial r} = 0 \quad \text{at } r = 0. \quad (62c)$$

On the other hand, for the DC model, the surface concentration is calculated using a correction from the closure problem,

$$\langle c_n \rangle_A^{\text{dc}} = \langle c_n \rangle^n + \langle s \rangle_A j_n^{\text{dc}}. \quad (63)$$

Note that this is *not* the closure problem presented in the main text, but rather that in the original publication of the DC model. The PDE governing volume-averaged concentration in the DC model is,

$$\partial_t \langle c_n \rangle^n = -\frac{A}{FV_n} j_n^{\text{dc}}. \quad (64)$$

In the DC model, ‘no-flux’ BCs are used at the electrode boundary with both the separator and the current collector,

$$\frac{\partial \langle c_n \rangle^n}{\partial x} = 0 \quad \text{at } x = L_s, \quad (65a)$$

$$\frac{\partial \langle c_n \rangle^n}{\partial x} = 0 \quad \text{at } x = L. \quad (65b)$$

The lengths of the separator and negative electrode are L_s and L_n respectively, and $L = L_s + L_n$.

2.2 Electrolyte Charge Balance

The remaining model equations are identical for DC and DFN. For charge conservation in the electrolyte, the PDE is,

$$\frac{\partial \bar{i}_e}{\partial x} = a_k j_k^m \quad \text{for } 0 < x < L. \quad (66)$$

where $\bar{i}_e$ is the macroscale electrolyte current,

$$\bar{i}_e = -\sigma_e \mathfrak{B}_e \frac{\partial \langle \phi_e \rangle^e}{\partial x} - \sigma_e \mathfrak{B}_e \theta \frac{\partial}{\partial x} \ln \langle c_e \rangle^e, \quad (67)$$

a_k is the specific surface area ($\frac{A}{V}$) and,

$$\mathfrak{B}_e = \begin{cases} \mathfrak{B}_{e,s} & \text{for } 0 \leq x \leq L_s, \\ \mathfrak{B}_{e,n} & \text{for } L_s < x \leq L, \end{cases} \quad a_k = \begin{cases} 0 & \text{for } 0 \leq x \leq L_s, \\ a_n & \text{for } L_s < x \leq L, \end{cases} \quad j_k^m = \begin{cases} 0 & \text{for } 0 \leq x \leq L_s, \\ j_n^m & \text{for } L_s < x \leq L, \end{cases} \quad (68)$$

where $\mathfrak{B}_e$ is the transport efficiency of the electrolyte phase, which is related to the porosity (ε_e) and tortuosity (τ_e) by,

$$\mathfrak{B}_e = \frac{\varepsilon_e}{\tau_e}. \quad (69)$$

The expression for reaction rate at the lithium foil is,

$$j_{\text{li}} = 2j_{\text{li},0} \sinh \left(-\frac{F}{2RT} \langle \phi_e \rangle^e \right). \quad (70)$$

For a constant-current discharge, the current density provided by the cell (i_{app}) will determine the current density at the lithium foil due to charge conservation. This expression can then be rearranged to find a Dirichlet BC for $\phi_{e,s}$. However, note that the porosity of the separator, $\varepsilon_{e,s}$, must be accounted for in this,

$$j_{\text{li}} = -\frac{i_{\text{app}}}{\varepsilon_{e,s}}, \quad (71)$$

and thus,

$$\langle \phi_e \rangle^e = -\frac{2RT}{F} \operatorname{arcsinh} \left(-\frac{i_{\text{app}}}{2\varepsilon_{e,s} j_{\text{li},0}} \right) \quad \text{at } x = 0. \quad (72)$$

For the BC at $x = L$, the CC is assumed to be impermeable to lithium ions. Using the expression for $\langle i_e \rangle^e$ with this statement we obtain,

$$\frac{\partial \langle \phi_e \rangle^e}{\partial x} = 0 \quad \text{at } x = L. \quad (73)$$

2.3 Electrolyte mass balance

The PDE is,

$$\varepsilon_e \partial_t \langle c_e \rangle^e = \frac{\partial}{\partial x} \left(D_e \mathfrak{B}_e \frac{\partial \langle c_e \rangle^e}{\partial x} \right) + F^{-1} (1 - t_+^0) a_n j_n^m \quad \text{for } 0 < x < L, \quad (74)$$

where the electrolyte volume fraction (ε_e) is

$$\varepsilon_e = \begin{cases} \varepsilon_{e,s} & \text{for } 0 \leq x \leq L_s, \\ \varepsilon_{e,n} & \text{for } L_s < x \leq L. \end{cases} \quad (75)$$

At the lithium-foil boundary with the effective separator, the macroscale electrolyte current ($\bar{i}_{e,s}$) and flux ($\bar{N}_{e,s}$) must balance the charge entering/leaving the cell at the CC,

$$i_{\text{app}} = \bar{i}_{e,s} \quad \text{at } x = 0, \quad (76a)$$

$$\frac{i_{\text{app}}}{F} = \bar{N}_{e,s} \quad \text{at } x = 0. \quad (76b)$$

The expression for $\bar{N}_e$,

$$\bar{N}_{e,s} = -\mathfrak{B}_{e,s} D_e \frac{\partial \langle c_e \rangle^e}{\partial x} + \frac{t_+^0}{F} \bar{i}_e, \quad (77)$$

can be used in Equation 76 to give

$$\frac{\partial \langle c_e \rangle^e}{\partial x} = -\frac{1}{F \mathfrak{B}_{e,s} D_e} (1 - t_+^0) i_{\text{app}} \quad \text{at } x = 0. \quad (78)$$

For the BC at $x = L$, the CC is assumed to be impermeable to lithium ions,

$$\frac{\partial \langle c_e \rangle^e}{\partial x} = 0 \quad \text{at } x = L. \quad (79)$$

Finally, we require in initial condition,

$$\langle c_e \rangle^e = c_{e,0} \quad \text{at } t = 0. \quad (80)$$

2.4 AM Charge Balance

The PDE is,

$$\frac{\partial}{\partial x} \left(\sigma_n \mathfrak{B}_n \frac{\partial \langle \phi_n \rangle^n}{\partial x} \right) = a_n j_n^m \quad \text{for } L_s < x < L. \quad (81)$$

where $\mathfrak{B}_n$ is the transport efficiency of the AM phase, which is related to AM volume fraction and tortuosity by,

$$\mathfrak{B}_n = \frac{\varepsilon_n}{\tau_n}. \quad (82)$$

An effective conductivity can also be defined as,

$$\sigma_n^{\text{eff}} = \sigma_n \mathfrak{B}_n. \quad (83)$$

For a galvanostatic (dis)charge, the current within the AM is fixed by the applied current density,

$$\frac{\partial \langle \phi_n \rangle^n}{\partial x} = -\frac{i_{\text{app}}}{\mathfrak{B}_n \sigma_n} \quad \text{at } x = L. \quad (84)$$

We assume that no reaction takes place at the interface between the electrode and the effective separator, giving

$$\frac{\partial \langle \phi_n \rangle^n}{\partial x} = 0 \quad \text{at } x = L_s. \quad (85)$$

3 External Boundary Conditions for the Closure Problem

3.1 Summary

To solve the closure problem derived in the main text (Equation 41), BCs for the external boundaries of Ω_e must be defined. Since the closure problem is solved across the whole of Ω_e and not a representative volume, the electrolyte meets the current collector ($\partial\Omega_{cc,e}$), outer walls ($\partial\Omega_{w,e}$), and a lithium foil ($\partial\Omega_{li,e}$). The BCs that appear at these interfaces are justified below, but can be summarised as,

$$\mathbf{n}_{e,w} \cdot \nabla b_x = 0 \quad \text{at } \partial\Omega_{w,e}, \quad (86a)$$

$$\mathbf{n}_{e,cc} \cdot \nabla b_x = 0 \quad \text{at } \partial\Omega_{cc,e}, \quad (86b)$$

$$\mathbf{n}_{e,li} \cdot \nabla b_x = 0 \quad \text{at } \partial\Omega_{li,e}. \quad (86c)$$

3.2 The Current Collector

The closure problem BC at the current collector is constrained by a macroscale BC. In the microscale model, we have the following constraints (Equations 57c and 59c)

$$\mathbf{n}_{e,cc} \cdot \mathbf{i}_e = 0 \quad \text{at } \partial\Omega_{cc,e}, \quad (87a)$$

$$\mathbf{n}_{e,cc} \cdot \mathbf{N}_e = 0 \quad \text{at } \partial\Omega_{cc,e}. \quad (87b)$$

Inserting the expressions for $\mathbf{N}_e$ and $\mathbf{i}_e$ (Equations 52a and 52b)

$$\mathbf{n}_{e,cc} \cdot (-\sigma_e \nabla \phi_e - \sigma_e \theta \nabla \ln c_e) = 0 \quad \text{at } \partial\Omega_{cc,e}, \quad (88a)$$

$$\mathbf{n}_{e,cc} \cdot (-D_e \nabla c_e) = 0 \quad \text{at } \partial\Omega_{cc,e}. \quad (88b)$$

Therefore

$$\mathbf{n}_{e,cc} \cdot (\nabla \phi_e) = 0 \quad \text{at } \partial\Omega_{cc,e}, \quad (89)$$

and since $\mathbf{n}_{e,cc}$ points parallel to the x axis

$$\partial_x \phi_e = 0 \quad \text{at } \partial\Omega_{cc,e}. \quad (90)$$

On the other hand, in the macroscale models we have the following constraint (Equation 73),

$$\partial_x \langle \phi_e \rangle^e = 0 \quad \text{at } \partial\Omega_{cc,e}. \quad (91)$$

Subtracting Equation 91 from Equation 90,

$$\partial_x \tilde{\phi}_e = 0 \quad \text{at } \partial\Omega_{cc,e}.$$

Inserting the expression for $\tilde{\phi}_e$ (Equation 40),

$$\partial_x b_x = 0 \quad \text{at } \partial\Omega_{cc,e}.$$

3.3 Outer Walls

Performing similar manipulations to those shown above for the current collector, the BC at $\partial\Omega_{w,e}$ in the microscale model (Equation 59a) can be rearranged to find,

$$\mathbf{n}_{e,w} \cdot \nabla \phi_e = 0 \quad \text{at } \partial\Omega_{w,e}. \quad (92)$$

Inserting the expression for ϕ_e (Equation 46),

$$\mathbf{n}_{e,w} \cdot \nabla (\langle \phi_e \rangle^e + b_x \partial_x \langle \phi_e \rangle^e) = 0 \quad \text{at } \partial\Omega_{w,e}. \quad (93)$$

As in the main text, $\partial_y \langle \phi_e \rangle^e$ and $\partial_z \langle \phi_e \rangle^e$ are negligible, leading to

$$\mathbf{n}_{e,w} \cdot \nabla b_x = -n_{e,w,x} \quad \text{at } \partial\Omega_{w,e}, \quad (94)$$

where $n_{e,w,x}$ is the x component of $\mathbf{n}_{e,w}$. However, $\mathbf{n}_{e,w}$ is perpendicular to the x axis (see Figure 12), and thus $n_{e,w,x} = 0$, giving

$$\mathbf{n}_{e,w} \cdot \nabla b_x = 0 \quad \text{at } \partial\Omega_{w,e}. \quad (95)$$

3.4 Lithium Foil

Deriving a BC at the lithium foil by comparison of microscale and upscaled models is less evident, as constraints only exist in integral form, e.g.,

$$\int_{\partial\Omega_{\text{li},e}} (\mathbf{n}_{e,\text{li}} \cdot \mathbf{i}_e) dA = i_{\text{app}}.$$

Therefore, to justify the BC at the lithium foil, we make an argument from the definition of effective properties. In the macroscale models (DFN or DC), the following is enforced (Equation 76),

$$i_{\text{app}} = \bar{i}_{e,s} \quad \text{at } \partial\Omega_{\text{li},e}, \quad (96a)$$

$$\frac{i_{\text{app}}}{F} = \bar{N}_{e,s} \quad \text{at } \partial\Omega_{\text{li},e}, \quad (96b)$$

which after inserting the expressions for $\bar{i}_{e,s}$ and $\bar{N}_{e,s}$ (Equations 67 and 77) gives

$$-\sigma_e \mathfrak{B}_{e,s} \partial_x \langle \phi_e \rangle^e - \sigma_e \mathfrak{B}_{e,s} \theta \partial_x \ln \langle c_e \rangle^e = i_{\text{app}} \quad \text{at } \partial\Omega_{\text{li},e}, \quad (97a)$$

$$-\mathfrak{B}_{e,s} D_e \partial_x \langle c_e \rangle^e = (1 - t_+^0) \frac{i_{\text{app}}}{F} \quad \text{at } \partial\Omega_{\text{li},e}. \quad (97b)$$

However, the effective transport efficiency in the separator, $\mathfrak{B}_{e,s}$, is defined by solving a transport problem which essentially enforces [54, 44, 55],

$$\int_{\partial\Omega_{\text{li},e}} (\mathbf{n}_{e,\text{li}} \cdot \nabla \phi_e) dA = \nabla \langle \phi_e \rangle^e \int_{\partial\Omega_{\text{li},e}} dA. \quad (98)$$

In other words, $\mathfrak{B}_{e,s}$ is defined with the objective of making macroscale potential gradients ($\nabla \langle \phi_e \rangle^e$) correspond as closely as possible to microscale potential gradients ($\nabla \phi_e$). Therefore, we make the assumption that

$$\mathbf{n}_{e,\text{li}} \cdot \nabla \phi_e = \mathbf{n}_{e,\text{li}} \cdot \nabla \langle \phi_e \rangle^e \quad \text{at } \partial\Omega_{\text{li},e}. \quad (99)$$

For the cases explored, the validity of this assumption is well supported by DNS results. Inserting the decomposition for ϕ_e ,

$$\mathbf{n}_{e,\text{li}} \cdot \nabla \tilde{\phi}_e = 0 \quad \text{at } \partial\Omega_{\text{li},e}, \quad (100)$$

and thus

$$\mathbf{n}_{e,\text{li}} \cdot \nabla b_x = 0 \quad \text{at } \partial\Omega_{\text{li},e}. \quad (101)$$

4 The Effect of the Reaction Source Term

4.1 Introduction

In the main text, the effect of the reaction rate on the $\tilde{\phi}_e$ field was neglected (see the developments directly after Equation 31). This was justified on the basis that $|j_n| \ll |\sigma_e \partial_x \langle \phi_e \rangle^e|$. Intuitively, one would assume that this assumption deteriorates at higher charging rates. However, whilst the magnitude of the reaction rate source term increases at higher C-rates, as does that of the potential gradients. For the 2C charge simulated in the main article (Electrode U), the DNS results reveal that $|j_n| = O(1) \text{ Am}^{-2}$, and $|\sigma_e \partial_x \langle \phi_e \rangle^e| = O(10^2) \text{ Am}^{-2}$, demonstrating that the above assumption remains reasonable. Nonetheless, we could advance an extended closure that avoids this assumption, which will be explored in this section.

4.2 Theory

To extend the closure to consider the effect of the reaction rate on the $\tilde{\phi}_e$ field, we will take inspiration from the development of the dual-continuum model published in our previous work [18–20]. In this, we focussed on the upscaling of the AM phase, and wrote the microscale model on a ‘per-particle’ basis. We can adopt a similar approach for the electrolyte phase, by decomposing the microscale electrolyte domain Ω_e into separate pores, Ω_{e_i} , where i is the pore index. The microscale equations on potential (Equation 14) now become,

$$-\nabla \cdot (\sigma_e \nabla \phi_{e_i}) - \nabla \cdot (\sigma_e \theta \nabla \ln c_{e_i}) = 0 \quad \text{in } \Omega_{e_i}, \quad (102a)$$

$$\mathbf{n} \cdot (-\sigma_e \nabla \phi_{e_i} - \sigma_e \theta \nabla \ln c_{e_i}) = -j_{n_i} \quad \text{at } \partial\Omega_{n,e_i}, \quad (102b)$$

whilst those for concentration are,

$$\partial_t c_{e_i} = \nabla \cdot (D_e \nabla c_{e_i}) \quad \text{in } \Omega_{e_i}, \quad (103a)$$

$$\mathbf{n} \cdot (D_e \nabla c_{e_i}) = F^{-1} (1 - t_+^0) j_{n_i} \quad \text{at } \partial\Omega_{n,e_i}. \quad (103b)$$

With this alternative form of the microscale problem, we then proceed through the developments identically to the main text, leading us to the following PDEs on $\tilde{\phi}_{e_i}$ (Equation 34) and $\tilde{c}_{e_i}$ (Equation 29),

$$\Delta \tilde{\phi}_{e_i} = \langle \Delta \tilde{\phi}_{e_i} \rangle^e \quad \text{in } \Omega_{e_i}, \quad (104a)$$

$$\Delta \tilde{c}_{e_i} = \langle \Delta \tilde{c}_{e_i} \rangle^e \quad \text{in } \Omega_{e_i}. \quad (104b)$$

In the BCs, we will no longer neglect the influence of the reaction rate, but instead approximate it using the surface-averaged value of reaction rate in that pore, $j_{n_i} \approx \langle j_{n_i} \rangle_A$, where

$$\langle j_{n_i} \rangle_A = \frac{1}{A_i} \int_{\mathcal{A}_i} j_{n_i} dA. \quad (105)$$

For $\tilde{\phi}_{e_i}$, the BC (Equation 30) becomes,

$$\mathbf{n} \cdot \left(-\sigma_e \nabla \langle \phi_{e_i} \rangle^e - \sigma_e \nabla \tilde{\phi}_{e_i} - \sigma_e \theta^* \nabla \langle c_{e_i} \rangle^e - \sigma_e \theta^* \nabla \tilde{c}_{e_i} \right) = -\langle j_{n_i} \rangle_A, \quad (106)$$

whilst for $\tilde{c}_{e_i}$ the BC (Equation 31) is now

$$\mathbf{n} \cdot (D_e \nabla \langle c_{e_i} \rangle^e + D_e \nabla \tilde{c}_{e_i}) = F^{-1} (1 - t_+^0) \langle j_{n_i} \rangle_A. \quad (107)$$

Equation 107 can be used to simplify Equation 106,

$$\mathbf{n} \cdot \left(-\nabla \langle \phi_{e_i} \rangle^e - \nabla \tilde{\phi}_{e_i} \right) = \langle j_{n_i} \rangle_A \left(\frac{\theta^*}{FD_e} (1 - t_+^0) - \frac{1}{\sigma_e} \right). \quad (108)$$

Given the source terms appearing in Equation 108, the closure proposed will now be,

$$\tilde{\phi}_{e_i} = \mathbf{b}_i \cdot \nabla \langle \phi_{e_i} \rangle^e + s_i \langle j_{n_i} \rangle_A, \quad (109)$$

where s_i is a new closure variable that incorporates the effect of surface reaction on the $\tilde{\phi}_e$ field. This closure will be simplified using an identical assumption to that in the main text: $\partial_y \langle \phi_e \rangle^e, \partial_z \langle \phi_e \rangle^e \approx 0$,

$$\tilde{\phi}_{e_i} = b_{x_i} \partial_x \langle \phi_{e_i} \rangle^e + s_i \langle j_{n_i} \rangle_A. \quad (110)$$

Inserting our expression for $\tilde{\phi}_{e_i}$ into the PDE (Equation 104a) and BC (Equation 108) gives

$$\Delta b_{x_i} \partial_x \langle \phi_{e_i} \rangle^e + \Delta s_i \langle j_{n_i} \rangle_A = \langle \Delta b_{x_i} \rangle^e \partial_x \langle \phi_{e_i} \rangle^e + \langle \Delta s_i \rangle^e \langle j_{n_i} \rangle_A \quad \text{in } \Omega_{e_i}, \quad (111a)$$

$$\mathbf{n} \cdot (-\nabla \langle \phi_{e_i} \rangle^e - \nabla b_{x_i} \partial_x \langle \phi_{e_i} \rangle^e) - \mathbf{n} \cdot \nabla s_i \langle j_{n_i} \rangle_A = \langle j_{n_i} \rangle_A \left(\frac{\theta^*}{FD_e} (1 - t_+^0) - \frac{1}{\sigma_e} \right) \quad \text{at } \partial\Omega_{n,e_i}. \quad (111b)$$

Now, the linearity of the equations is exploited to arrive at two independent closure problems,

$$\Delta b_{x_i} = \langle \Delta b_{x_i} \rangle^e \quad \text{in } \Omega_{e_i}, \quad (112a)$$

$$\mathbf{n} \cdot \nabla b_{x_i} = -n_x \quad \text{at } \partial\Omega_{n,e_i}, \quad (112b)$$

and

$$\Delta s_i = \langle \Delta s_i \rangle^e \quad \text{in } \Omega_{e_i}, \quad (113a)$$

$$\mathbf{n} \cdot \nabla s_i = - \left(\frac{\theta^*}{FD_e} (1 - t_+^0) - \frac{1}{\sigma_e} \right) \quad \text{at } \partial\Omega_{n,e_i}. \quad (113b)$$

Equation 112 is identical to the closure in the main article (Equation 41), whilst Equation 113 is a new closure problem. The latter is solvable in its current form. However several manipulations make its solution more practical. Firstly, the Divergence Theorem gives the following relationship,

$$\langle \Delta s_i \rangle^e = \frac{1}{V_{e_i}} \int_{\mathcal{A}_i} \mathbf{n} \cdot \nabla s_i dA, \quad (114)$$

into which Equation 113b can be inserted to find

$$\langle \Delta s_i \rangle^e = - \left(\frac{\theta^*}{FD_e} (1 - t_+^0) - \frac{1}{\sigma_e} \right) \frac{A_i}{V_{e_i}}. \quad (115)$$

This can be used to rewrite the PDE (Equation 113a). Finally, using the following dimensionless groups,

$$s_i = L \left(\frac{\theta^*}{FD_e} (1 - t_+^0) - \frac{1}{\sigma_e} \right) \hat{s}_{\phi,i}, \quad (116a)$$

$$x = \hat{x}L. \quad (116b)$$

the problem can be cast in dimensionless form,

$$\hat{\Delta} \hat{s}_i = - \frac{LA_i}{V_{e_i}} \quad \text{in } \Omega_{e_i}, \quad (117a)$$

$$\mathbf{n} \cdot \hat{\nabla} \hat{s}_i = -1 \quad \text{at } \partial\Omega_{n,e_i}. \quad (117b)$$

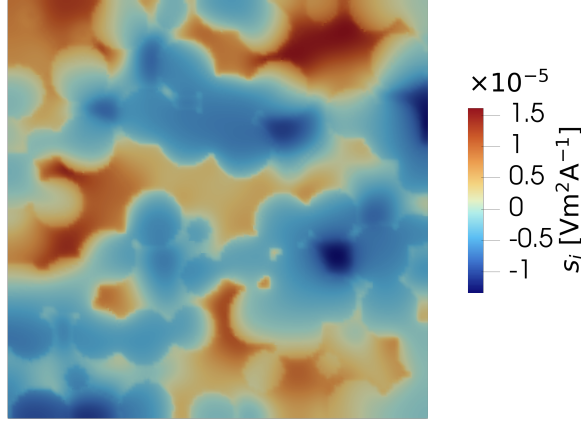


Figure 13: The closure variable solution, s_i , over the interface between the negative electrode and the separator regions.

4.3 Results and Discussion

The practical advantage of Equation 117 is that it can be solved using the open-source OpenFOAM tool [45] developed in our previous work, as it takes the exact same form as the problem tackled in our earlier publications [18–20]. To explore the magnitude of additional terms in this extended closure problem, we reconsider Electrode U from the main article. We solve both closure problems presented above (Equations 112 and 117), and compare the magnitude of the contributions to $\tilde{\phi}_e$,

$$\tilde{\phi}_{e_i} = \underbrace{b_{x_i} \partial_x \langle \phi_{e_i} \rangle^e}_{\text{Term 1}} + \underbrace{s_i \langle j_{n_i} \rangle_A}_{\text{Term 2}}. \quad (118)$$

Values of $\partial_x \langle \phi_{e_i} \rangle^e$ and $\langle j_{n_i} \rangle_A$ were taken from the DC model results, as in the main article. To identify the pore boundaries in Electrode U – which is required for solving Equation 117 – the watershed algorithm provided in our open-source tool [45] was used (with default settings).

The solution to the additional closure problem, s_i , is shown in Figure 13. Results for the 2 C charge are shown in Figure 14. Here, Term 1 is roughly two orders of magnitude greater than Term 2. Thus, perhaps surprisingly, the macroscale electrolyte potential gradients have a significantly larger impact on the $\tilde{\phi}_{e_i}$ fields compared to the surface reaction, even at relatively high C-rates. This demonstrates that neglecting the surface-reaction source terms (as in the main text) is a reasonable approach for capturing the most influential phenomena in the closure problem. This also supports the argument advanced in the main article: neglecting the surface-reaction source terms is not a primary cause of the errors in reconstructing overpotential (see Figures 5 and 7), which were instead attributed to difficulties in estimating $\partial_x \langle \phi_e \rangle^e$ with upscaled models.

5 Extending the Closure Expression

In the main text we noted that a more complex expression could be used for $\tilde{\phi}_e$, but that it would eventually reduce to Equation 35 due to the assumptions already introduced – this is demonstrated here. Given the source terms in Equations 27 and 32, the closure expressions are extended to

$$\tilde{\phi}_e = \mathbf{b}_{\phi\phi} \cdot \nabla \langle \phi_e \rangle^e + \mathbf{b}_{\phi c} \cdot \nabla \langle c_e \rangle^e, \quad (119a)$$

$$\tilde{c}_e = \mathbf{b}_{cc} \cdot \nabla \langle c_e \rangle^e + \mathbf{b}_{c\phi} \cdot \nabla \langle \phi_e \rangle^e. \quad (119b)$$

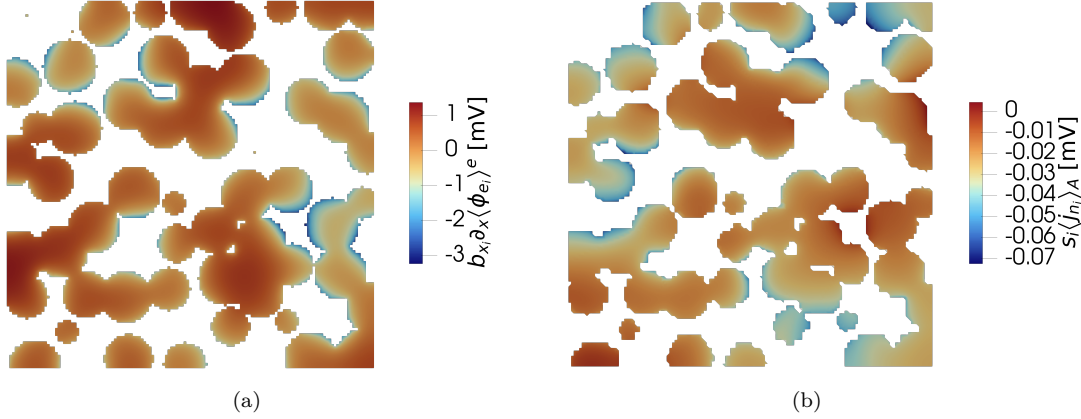


Figure 14: The $\tilde{\phi}_e$ contributions for a 2C charge from the $\partial_x \langle \phi_{ei} \rangle^e$ source term considered in the main article (a), and the reaction source term ($\langle j_{ni} \rangle_A$) which was neglected in the main text (b). The distortions in the geometry of Figure b are an artifact of Paraview's mesh re-sampling algorithm.

The problem for $\tilde{\phi}_e$ will be treated first. Inserting Equation 119a into Equation 34 gives

$$\begin{aligned} \Delta \mathbf{b}_{\phi\phi} \cdot \nabla \langle \phi_e \rangle^e + \Delta \mathbf{b}_{\phi c} \cdot \nabla \langle c_e \rangle^e &= \langle \Delta \mathbf{b}_{\phi\phi} \cdot \nabla \langle \phi_e \rangle^e + \Delta \mathbf{b}_{\phi c} \cdot \nabla \langle c_e \rangle^e \rangle^e & \text{in } \Omega_e, & (120a) \\ \mathbf{n} \cdot (\nabla \mathbf{b}_{\phi\phi} \cdot \nabla \langle \phi_e \rangle^e + \nabla \mathbf{b}_{\phi c} \cdot \nabla \langle c_e \rangle^e) &= -\mathbf{n} \cdot \nabla \langle \phi_e \rangle^e & \text{at } \partial\Omega_{n,e}. & (120b) \end{aligned}$$

By exploiting the linearity of the problem, we decompose this into

$$\begin{aligned} \Delta \mathbf{b}_{\phi\phi} &= \langle \Delta \mathbf{b}_{\phi\phi} \rangle^e & \text{in } \Omega_e, & (121a) \\ \mathbf{n} \cdot \nabla \mathbf{b}_{\phi\phi} &= -\mathbf{n} & \text{at } \partial\Omega_{n,e}, & (121b) \end{aligned}$$

and

$$\begin{aligned} \Delta \mathbf{b}_{\phi c} &= \langle \Delta \mathbf{b}_{\phi c} \rangle^e & \text{in } \Omega_e, & (122a) \\ \mathbf{n} \cdot \nabla \mathbf{b}_{\phi c} &= 0 & \text{at } \partial\Omega_{n,e}, & (122b) \end{aligned}$$

where $\nabla \langle \phi_e \rangle^e$ and $\nabla \langle c_e \rangle^e$ have been removed from the averaging operator as in the main text. By manipulation with the divergence theorem,

$$\langle \Delta \mathbf{b}_{\phi c} \rangle^e = \frac{1}{V_e} \int_A (\mathbf{n} \cdot \nabla \mathbf{b}_{\phi c}) dA, \quad (123)$$

into which Equation 122b can be inserted to give

$$\langle \Delta \mathbf{b}_{\phi c} \rangle^e = 0. \quad (124)$$

Since the BCs at the external boundaries of Ω_e also take the form $\mathbf{n} \cdot \nabla \mathbf{b}_{\phi c} = 0$, the only solution to Equation 122 is $\mathbf{b}_{\phi c} = 0$, and thus

$$\tilde{\phi}_e = \mathbf{b}_{\phi\phi} \cdot \nabla \langle \phi_e \rangle^e. \quad (125)$$

The problem for $\tilde{c}_e$ will now be treated. Inserting Equation 119b into Equations 29 and 33,

$$\begin{aligned} \Delta \mathbf{b}_{cc} \cdot \nabla \langle c_e \rangle^e + \Delta \mathbf{b}_{c\phi} \cdot \nabla \langle \phi_e \rangle^e &= \langle \Delta \mathbf{b}_{cc} \cdot \nabla \langle c_e \rangle^e + \Delta \mathbf{b}_{c\phi} \cdot \nabla \langle \phi_e \rangle^e \rangle^e & \text{in } \Omega_e, & (126a) \\ \mathbf{n} \cdot (\nabla \mathbf{b}_{cc} \cdot \nabla \langle c_e \rangle^e + \nabla \mathbf{b}_{c\phi} \cdot \nabla \langle \phi_e \rangle^e) &= -\mathbf{n} \cdot \nabla \langle c_e \rangle^e & \text{at } \partial\Omega_{n,e}. & (126b) \end{aligned}$$

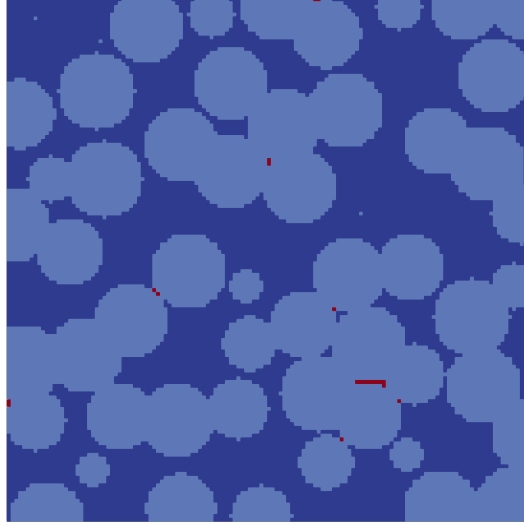


Figure 15: Electrode U microstructure at the separator interface from Figure 5 with inactive material in red.

As above, we exploit the linearity of the equations and decompose into two closure problems,

$$\begin{aligned} \Delta \mathbf{b}_{cc} &= \langle \Delta \mathbf{b}_{cc} \rangle^e & \text{in } \Omega_e, & (127a) \\ \mathbf{n} \cdot \nabla \mathbf{b}_{cc} &= -\mathbf{n} & \text{at } \partial\Omega_{n,e}, & (127b) \end{aligned}$$

and

$$\begin{aligned} \Delta \mathbf{b}_{c\phi} &= \langle \Delta \mathbf{b}_{c\phi} \rangle^e & \text{in } \Omega_e, & (128a) \\ \mathbf{n} \cdot \nabla \mathbf{b}_{c\phi} &= 0 & \text{at } \partial\Omega_{n,e}. & (128b) \end{aligned}$$

Upon inspection, Equation 127 is clearly identical to Equation 121, and Equation 128 is identical to Equation 122. Therefore,

$$\tilde{\phi}_e = \mathbf{b} \cdot \nabla \langle \phi_e \rangle^e, \quad (129a)$$

$$\tilde{c}_e = \mathbf{b} \cdot \nabla \langle c_e \rangle^e, \quad (129b)$$

where $\mathbf{b} = \mathbf{b}_{cc} = \mathbf{b}_{\phi\phi}$.

6 Apparent Differences in the DNS and Closure Microstructures

In the main article, Figures 5 and 7 suggest very small differences between the microstructures used for the DNS and the for the closure problem. The microstructures are in fact identical. This observation results from a very small percentage of inactive material being present in the structures, which is used to replace any isolated electrolyte regions – see Figure 15.

7 Solver Parameters

For the macroscale models, a discretisation of 20 nodes in the separator, 40 nodes in the negative electrode, and 60 nodes in the radial direction (only relevant for the DFN) was used. PyBaMM's

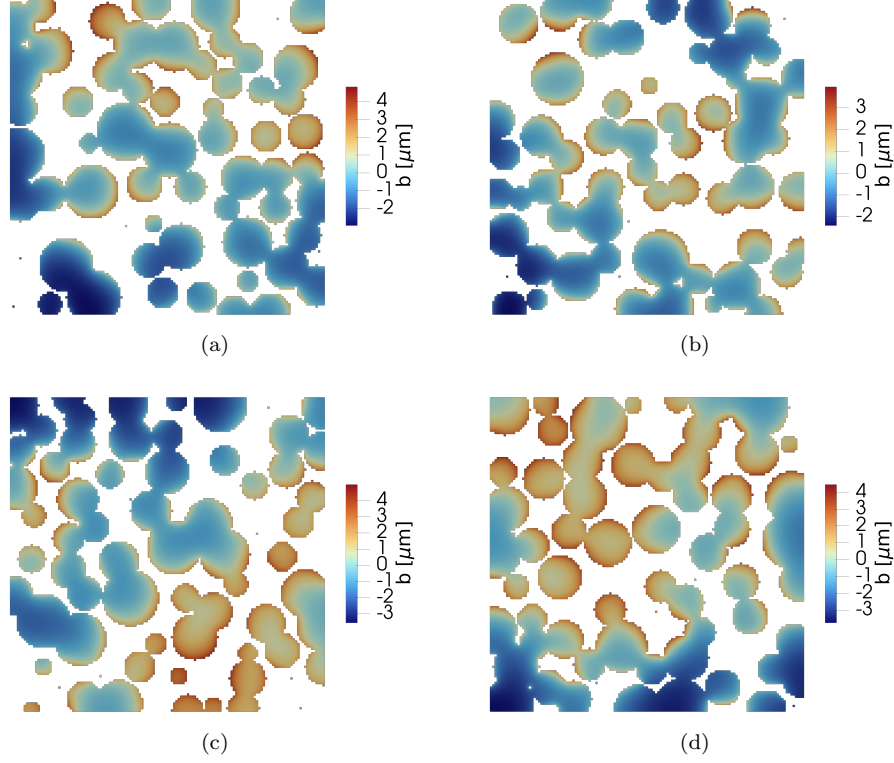


Figure 16: Solutions to b_x across the separator interface for Electrodes 1 (a), 2 (b), 3 (c), and 4 (d).

IDAKLU solver was used with default parameters to solve all models. Computations were performed using a single core on a Windows Subsystem for Linux system with a 12th Gen Intel® Core™ i7-12700H 2.30 GHz processor and 32 GB of RAM available.

8 Reconstructing Lithium-Plating Overpotential

To reconstruct η_{pl}^r , one requires solutions to the closure problem (b_x) and an upscaled model (to obtain $\partial_x \langle \phi_e \rangle^e$, $\langle \phi_e \rangle^e$, and $\langle \phi_n \rangle^n$). The solutions to b_x across the separator interface for Electrodes 1, 2, 3, and 4 are shown in Figure 16. Likewise, the values of $\partial_x \langle \phi_e \rangle^e$, $\langle \phi_e \rangle^e$, and $\langle \phi_n \rangle^n$ used are shown in Table 5.

Table 5: Values required for reconstructing η_{pl}^r (taken from the DC model at the point of charge termination).

Electrode	$\partial_x \langle \phi_e \rangle^e \text{ [Vm}^{-1}\text{]}$	$\langle \phi_e \rangle^e \text{ [V]}$	$\langle \phi_n \rangle^n \text{ [V]}$
U	-552.4	-0.10190	-0.10054
1	-538.1	-0.10186	-0.10027
2	-552.8	-0.10178	-0.10045
3	-541.2	-0.10183	-0.09990
4	-551.1	-0.10180	-0.09973
R	-131.4	-0.04230	-0.04174